
IAL Biology Unit 3

WBI13

Practical Skills in Biology I

A* revision and exam-technique guide

Built from analysis of 21 past papers and mark schemes,
May 2019 to May 2026

WBI13/01 · 50 marks · 1 hour 20 minutes
20% of the IAS · 10% of the full IAL

1 · What the papers actually look like

Unit 3 is structurally unlike Units 1 and 2, and preparing for it as though it were a content paper is the main reason candidates underperform on it.

- **Three questions. 50 marks. 80 minutes.** Every one of the 21 papers.
- **No mark ramp.** Q1 averaged 15.8 marks, Q2 16.7, Q3 17.5. All three are equally weighted, so there is no "easy start" and no reason to rush the early pages.
- **Each question is a single unfamiliar investigation** — a method you have never seen, usually set out as a flow chart, followed by sub-parts probing the design, the data and the conclusions.
- **You are not expected to recognise the experiment.** The biology is nearly always Unit 1 or Unit 2 content in an unfamiliar wrapper. What is being marked is whether you can think like an experimenter.

TIMING — THIS PAPER GIVES YOU MORE TIME THAN YOU THINK

50 marks in 80 minutes is **1.6 minutes per mark**, against roughly 1.1 in Units 1 and 2. Budget about **24 minutes per question** and keep 8 minutes back. Use the surplus on the graph: a carefully drawn graph is four marks and takes six or seven minutes to do properly. Candidates who rush it to "save time" lose more than they save.

Command words, counted across all 21 papers

Command	Per paper	What it means here
Describe	3.6	The dominant command — unusual, and specific to this paper. Usually "describe how to..." a procedure: make a solution, measure the dependent variable, process the results. See §4.
Explain	2.9	Almost always "explain why" a design decision was taken — why a variable was controlled, why that temperature, why a control was included, why initial rate was used. See §3.
Calculate	2.3	Percentage change, rate, ratio, mitotic index, magnification, tensile strength. See §5.
Draw / Plot	2.5 combined	Drawing a graph or a results table. Marked against named criteria — §2 is the highest-value section in this guide.
State / Give / Name	4.0 combined	Variables, hazards, apparatus. One word or phrase.
Suggest	1.1	Unfamiliar by design. Reason from principles; never leave blank.
Comment / Evaluate / Justify / Discuss	1.3 combined	Conclusions and limitations. Needs evidence and a judgement. See §6.

Skill frequency across the 21 papers

Skill area	Papers (of 21)	Verdict
Graphs — plotting and interpretation	21	Every paper. These three are the spine of the qualification.
Calculations and data processing	21	
Describing or devising a method	20	
Variables — independent, dependent, control	14	Very likely.
Repeats, means, anomalies, reliability	10	
Conclusions and evaluation	10	
Aseptic and microbiology technique	7	Periodic , but each is a self-contained block of technique worth learning in an hour.
Microscopy and measurement (graticule)	5	
Statistics — SD, error bars, significance	5	
Serial dilution and making solutions	4	
Risk assessment, hazards, ethics	1 each	Rare — but each costs ten minutes to prepare. Do not drop them entirely.

THE FINDING THAT SHOULD CHANGE HOW YOU REVISE

Formal statistical tests are barely used in this paper. Across 21 papers, "t-test" appeared in **none**, "correlation coefficient" in **none**, chi-squared in two. **Standard deviation** appeared in eight. Yet "significant" appeared **37 times** across the mark schemes.

What that means: you are not asked to **compute** a test. You are asked to **interpret spread and significance** — usually from standard deviations or error bars. Learn that interpretation properly (§5.3) and treat the mechanics of t-tests as low priority.

The biological contexts the papers use

You cannot predict the context, but the pool is narrower than it feels. Across the 21 papers: plant tissue in sucrose or salt solutions — potato, carrot, beetroot — appeared in **12**; enzyme investigations in **6**; seed germination in **6**; eyepiece graticule measurement in **6**; vitamin C and DCPIP in **5**; plant minerals in **5**; antimicrobial zones of inhibition in **4**; tensile strength of fibres in **4**; mitotic index from root tips in **3**. Section 4 covers the techniques behind all of these.

2 · The graph and table marks

This is the single most reliable block of marks on the paper, and it is awarded against a fixed set of criteria that the mark schemes label with letters. Learn the letters and you can self-mark every graph you draw in practice.

THE GRAPH MARK SCHEME, AS IT ACTUALLY APPEARS

Graph questions are typically worth **four marks**, one for each criterion:

Code	Criterion	What earns it
A	Axes correct	The independent variable on the x-axis and the dependent variable on the y-axis . Where required, the axis starts at zero with no break in it.
L	Labels	Both axes labelled, with units . Copy the label and unit exactly as they appear in the table of results.
P	Plotting	All points plotted accurately , on suitable linear scales . Scales must not be awkward (avoid 3s and 7s) and should use at least half the grid.
S	Scale / line / chart type	Depends on the question: a suitable line of best fit , or the correct bar chart format, or a suitable linear scale starting at zero with no interruption .
K	Key	Awarded when there is more than one data set — a key or clear labels distinguishing them.

2.1 Choosing the right graph — the decision that carries the S mark

If the independent variable is...	Draw	Notes
Continuous (concentration, time, temperature, pH)	Line graph with a line of best fit	A line of best fit is a smooth curve or straight line with roughly equal numbers of points either side. Not dot-to-dot — the mark schemes say so explicitly.
Categoric (species, mineral missing, type of extract)	Bar chart with gaps between bars	Bars of equal width, separated by equal gaps. The y-axis must start at zero.
Continuous and grouped into classes	Histogram , no gaps between bars	Bars touch, because the data is continuous.
Two measured variables, looking for a relationship	Scatter graph	Used when you are testing for correlation rather than for an effect.

GRAPH ERRORS THE MARK SCHEMES NAME

Dot-to-dot instead of a line of best fit · axes the wrong way round · a broken or interrupted axis where one is not allowed · missing units · a scale that uses less than half the grid · no key when two data sets are plotted · plotting the **raw** data when the table gives a **mean**, or the reverse.

2.2 Drawing a results table

"Draw a suitable table to show these results" appeared repeatedly. The conventions:

- The **independent variable in the first (left-hand) column**; the dependent variable in the columns to its right.
- **Headings with units**, and units given in the **heading only** — never repeated beside each value in the body of the table.
- Columns for each **repeat**, plus a column for the **mean**, and where relevant a column for the processed value (percentage change, rate).
- All values in a column to the **same number of decimal places**, consistent with the resolution of the instrument used.
- Ruled lines, full borders, and no data left in the headings.

3 · Experimental design — the "explain why" marks

Nearly three "explain why" questions per paper, and they are almost always about a design decision. The answers follow a small number of templates.

3.1 Variables

Term	Definition and how to identify it
Independent variable	The variable the investigator deliberately changes . In a flow chart it is the thing that differs between the treatments.
Dependent variable	The variable that is measured . Ask what number ends up in the results table.
Control variables	Variables kept constant so that any change in the dependent variable can be attributed to the independent variable alone.

"EXPLAIN WHY X WAS KEPT CONSTANT" — THE TWO-PART TEMPLATE

1. Say what X would otherwise affect, with the mechanism — "temperature affects the **kinetic energy** of the molecules and therefore the rate of enzyme activity".
2. Say why that matters — "so if it varied, any change in the dependent variable could **not be attributed to the independent variable**, and the results would not be valid".

Naming the variable alone scores nothing. The mark is for the **consequence**. The phrase "so the effect of the independent variable can be isolated" is worth memorising.

Control variables that recur, with the reason each matters:

Variable	Why it must be controlled
Temperature	Affects kinetic energy, so rate of reaction, enzyme activity, diffusion and metabolism
pH	Away from the optimum, ionic and hydrogen bonds in the tertiary structure are disrupted, so the active site changes shape
Enzyme and substrate concentration	Both affect the number of enzyme-substrate complexes formed per unit time
Volume of solution	Affects concentration, and the surface area of tissue in contact with it
Size / surface area / mass of tissue	Affects the rate of diffusion into or out of the tissue
Time	Longer exposure gives a larger change, independent of the treatment
Age, variety or source of the organism	Genetic and physiological differences affect the response

3.2 Control experiments

A control is **not** the same as a control variable. A control experiment is an additional treatment run **without** the factor being tested.

THE STANDARD ANSWER

"The discs were soaked in the **solvent alone**, with no extract. This shows that any zone of inhibition is due to the **extract itself** and **not to the solvent**, giving a baseline for comparison."

Transfer the shape to any context: the control shows that the observed effect is **caused by the independent variable and not by something else in the procedure**. Where living material is used, a control may be boiled or denatured enzyme, or distilled water in place of the test solution.

3.3 Why that temperature, that pH, that duration

Question	Answer
Why 25 °C / 30 °C?	It is close to the optimum for the organism or enzyme, giving a measurable rate, while being warm enough to be above room temperature so the rate is not too slow. In school microbiology, incubating at 25 °C rather than 37 °C reduces the risk of culturing human pathogens .
Why was pH maintained?	pH affects the tertiary structure and therefore the shape of the enzyme's active site, so it must be held constant for the effect of the independent variable to be isolated. A buffer solution is used to maintain it.
Why measure the initial rate?	At the start, substrate concentration is highest and is not yet limiting, so the rate is at its maximum and reflects the conditions set. As the reaction proceeds, substrate is used up and the rate falls, so a later measurement would depend on how long you waited and would not be a fair comparison between treatments.
Why incubate for a fixed time?	So that time is a control variable and all treatments are compared over the same period.
Why leave discs for the solvent to	So that the solvent itself does not inhibit the bacteria and affect the zone of inhibition.

3.4 Repeats, means and anomalies

- **Why repeat?** To identify **anomalous results**, to allow a **mean** to be calculated which reduces the effect of **random error**, and to make the results more **reliable**.
- **Anomalous results** should be **identified and excluded from the mean**, not silently ignored. State that you would exclude them and say why.
- More repeats also allow a **standard deviation** to be calculated, which indicates the **spread** of the data.

FIVE TERMS EXAMINERS EXPECT YOU TO USE CORRECTLY

Accurate — close to the true value.

Precise — repeat measurements are close to one another. Data can be precise but inaccurate.

Resolution — the smallest change an instrument can detect (a balance reading to 0.01 g has higher resolution than one reading to 0.1 g).

Reliable / reproducible — the same result is obtained on repetition.

Valid — the experiment actually measures what it set out to measure, with all other variables controlled. An invalid experiment cannot be rescued by repeating it.

4 · Describing a method — the top command word

3.6 "describe" questions per paper. They are marked point by point, so write in **numbered steps**, one instruction per line, each containing a specific quantity or piece of apparatus. Vague verbs score nothing: "measure the mass" earns a mark, "see how heavy it is" does not.

THE STRUCTURE THAT SCORES ON ANY "DESCRIBE HOW TO..." QUESTION

1. What apparatus, with the measurement and its unit.
2. What quantity or concentration, stated numerically.
3. How the independent variable is varied, and over what range.
4. How the dependent variable is measured, and when.
5. Which variables are controlled.
6. Repeats and means.

4.1 Making a solution of known concentration

1. Weigh a known mass of the solid on a balance — state the mass.
2. Dissolve it in **less than** the final volume of distilled water, stirring.
3. Transfer to a **volumetric flask** and make up to the mark with distilled water, so that the final volume is known exactly.
4. **Filter** the solution if solid remains, to give a clear solution — important where the dependent variable is measured by colour or absorbance.

4.2 Serial dilution

To make a range of concentrations from one stock solution: take **1 cm³ of the stock** and add **9 cm³ of distilled water** to give a **1 in 10** dilution. Mix, then take 1 cm³ of that and repeat. Each step dilutes by the same factor, giving a geometric series. State the factor and show one calculation.

4.3 Aseptic technique

- **Sterilise the inoculating loop** by heating it in a Bunsen flame until red hot, and allow it to cool.
- Work **close to the Bunsen flame**, which creates an updraught carrying airborne microbes away.
- Lift the lid of the Petri dish **only slightly, and at an angle** — do not remove it.
- **Flame the neck** of culture bottles before and after pouring.
- Secure the lid with tape but **do not seal it completely**, so that **anaerobic** pathogens cannot grow.
- Incubate at **no more than 25 °C** in a school laboratory, to reduce the risk of growing organisms pathogenic to humans.
- Store dishes **inverted** so condensation does not drip onto the agar.

Zone of inhibition: measure the **diameter** of the clear area in mm with a ruler, taking two readings at right angles and meaning them; a larger zone indicates a more effective antimicrobial. Convert to area with πr^2 if the question asks.

4.4 Microscopy and the eyepiece graticule

CALIBRATION — THE SEQUENCE THAT EARNS THE MARKS

1. Place a **stage micrometer** on the stage; it has a scale of known dimensions.
2. Focus and **line up the eyepiece graticule scale against the stage micrometer scale**.
3. Count how many **eyepiece divisions** correspond to a known length on the stage micrometer.
4. Divide to find the length represented by **one eyepiece division** at that magnification.
5. Remove the micrometer, view the specimen, count eyepiece divisions and multiply.

The calibration is **only valid for that objective lens** — changing magnification requires recalibration. That point is often the mark.

Magnification = image size ÷ actual size. Convert units before dividing: 1 mm = 1000 µm = 1 000 000 nm.

4.5 The recurring plant-tissue investigation

Potato, carrot or beetroot cylinders in a range of sucrose or salt concentrations appeared in 12 of 21 papers. The method and the reasoning behind it:

- Cut cylinders with a **cork borer** so the diameter is constant, then cut to **equal length** with a scalpel against a ruler.
- **Blot dry** before weighing, both before and after, to remove surface liquid that would otherwise add to the mass.
- Leave for a fixed time in a known concentration; keep temperature constant.
- Calculate **percentage change in mass**, not raw change, **because the initial masses differ** — percentage change allows a fair comparison. This is a very common 2-mark explanation.
- Plot percentage change against concentration; the concentration where the line **crosses zero** is the concentration at which there is **no net movement of water**, so the water potential of the solution equals that of the tissue.

For **beetroot membrane permeability**, the dependent variable is the amount of red pigment released, measured with a **colorimeter** as absorbance. Rinse the cut discs first to remove pigment released by cutting — otherwise the baseline is wrong.

4.6 Other recurring techniques

Investigation	What to know
Vitamin C with DCPIP	DCPIP is blue and becomes colourless when reduced by vitamin C. Titrate the sample into a fixed volume of DCPIP until the blue colour just disappears. Compare the volume needed against a standard vitamin C solution of known concentration — the smaller the volume needed, the higher the vitamin C content.
Mitotic index from a root tip	Hydrolyse the root tip in acid, stain (e.g. with acetic orcein or toluidine blue), squash under a coverslip to produce a single layer of cells, and count. Mitotic index = cells in mitosis ÷ total cells . Count several fields of view and only cells where the stage is clear.
Tensile strength of plant fibres	Hang increasing masses from a fibre of known length and diameter until it breaks. Tensile strength = force ÷ cross-sectional area , with cross-sectional area = πr^2 . Control the fibre length, diameter and how the mass is added.
Seed germination	Equal numbers of seeds on moist filter paper, same temperature, same light conditions, same volume of water. Dependent variable is percentage germination or mean radicle length after a fixed time.
Plant mineral deficiency	Grow plants in culture solutions each lacking one named mineral, with a complete solution as the control. Compare height, mass or leaf colour after a fixed period.

5 · Calculations and data processing

In all 21 papers, averaging 2.3 per paper. Taken from the actual papers:

Type	Method	The trap
Percentage change	$(\text{final} - \text{initial}) \div \text{initial} \times 100$. Keep the sign — a negative value means a decrease.	Dividing by the final value; dropping the minus sign
Rate of change	change $\div$ time, with units such as g day ⁻¹ . From a curve, draw a tangent and take its gradient.	Omitting units; reading the gradient from the wrong points
Initial rate	Gradient of the tangent at t = 0 .	Using the overall gradient instead of the tangent at the start
Mitotic index	cells in mitosis $\div$ total cells ($\times 100$ for a percentage)	Counting interphase cells as dividing
Magnification	image size $\div$ actual size; convert units first	Not converting mm to μm before dividing
Tensile strength	force $\div$ cross-sectional area; area = πr^2 , and r is half the diameter	Using diameter in place of radius
Ratio	Divide both by the smaller; give in the order asked; simplify	Reversing the order
Dilution factor	Final volume $\div$ volume of stock used	Forgetting that 1 cm ³ into 9 cm ³ of water gives a total of 10 cm ³
Area of a zone of inhibition	πr^2 , where the measured value is usually the diameter	Substituting the diameter for r

FOUR RULES

1. Show every step — method marks survive an arithmetic slip, and the mark schemes award them. **2.** Give units, and the significant figures or decimal places asked for. **3.** Convert units **before** calculating. **4.** Check the magnitude is biologically sensible.

5.1 Significant figures and decimal places

Give a processed value to the **same number of significant figures as the least precise piece of raw data** used in the calculation. Do not copy out a calculator display to eight figures — this is explicitly penalised. Where the question specifies, follow the instruction exactly.

5.2 Uncertainty and percentage error

The uncertainty of a single reading is normally **half the smallest division** of the instrument. For a measurement requiring two readings — such as a length measured from both ends, or a change in mass — the uncertainty is **doubled**.

percentage error = (uncertainty $\div$ measured value) $\times 100$

The consequence worth stating in an evaluation: **percentage error is largest for the smallest measurements**. So using larger volumes, longer times or greater masses reduces percentage error — a standard improvement to suggest.

5.3 Standard deviation, error bars and significance

This is where the real statistics marks are, and the mark schemes return to it constantly — "significant" appeared 37 times.

- **Standard deviation** measures the **spread of data about the mean**. A **large** SD means widely spread data and less confidence in the mean; a **small** SD means the data are clustered closely and the mean is more representative.
- **Error bars** drawn at $\pm$ one standard deviation show that spread graphically.

INTERPRETING ERROR BARS — THE EXACT WORDING

If the error bars **overlap**, the difference between the means is **probably not significant**, and may be due to chance.

If the error bars **do not overlap**, the difference is **likely to be significant**.

Say "likely" and "probably" — the bars **suggest** significance, they do not prove it. A statistical test would be needed to establish it. That qualification is frequently the final mark.

Which test for what, in case it is asked: **Student's t-test** compares the means of two groups. **Chi-squared** compares observed with expected frequencies. **Correlation coefficient** measures the strength of a relationship between two variables. A result is conventionally taken as significant when the probability that it arose by chance is **less than 0.05 (5%)**, which means the **calculated value exceeds the critical value** at the appropriate degrees of freedom.

6 · Conclusions, limitations and improvements

6.1 Drawing a conclusion

- State the relationship, **using figures from the data with units**. A conclusion with no numbers in it rarely gets full marks.
- Answer the **original question** the investigation set out to address.
- Stay **within what the data shows** — do not extrapolate beyond the range tested, and do not generalise to other species or conditions that were not investigated.
- Where the data shows an association, say **correlation, not causation**, and say what further work would be needed.

6.2 Limitations — the ones that apply almost anywhere

Limitation	Improvement to pair it with
Too few repeats, so anomalies cannot be identified and the mean is unreliable	Increase the number of repeats and calculate a mean and standard deviation
Small or unrepresentative sample	Use a larger sample; select randomly
A variable was not controlled	Name it and say how to control it — a water bath, a buffer, a fixed time
Measuring instrument of low resolution	Use an instrument with finer graduations, or a data logger
Subjective judgement of an end point — a colour change judged by eye	Use a colorimeter , or a colour standard for comparison
Only a narrow range of the independent variable tested	Extend the range, and use smaller intervals around the point of interest
Biological variation between organisms or tissue samples	Use material from the same source, same age and same variety

An evaluation that lists limitations without **pairing each with a specific improvement** does not reach the top band. "Do more repeats" on its own is worth little; "do five repeats at each concentration and calculate a mean, so anomalies can be identified" is worth the mark.

6.3 Risk assessment and ethics

Rare — one paper each — but cheap to prepare. A risk assessment answer must name **the hazard, the risk it poses, and the precaution**:

- Scalpel or cork borer — cuts — cut downwards onto a tile, away from the hand.
- Microorganisms — infection — aseptic technique, incubate at no more than 25 °C, do not open sealed plates.
- Ethanol — flammable — keep away from naked flames.
- Acids, alkalis, DCPIP, stains — irritant to eyes and skin — wear goggles and gloves.
- Hot water baths — scalds — handle with tongs.

For work with living organisms, the ethical points are **minimising harm and distress**, using the **smallest number of organisms** necessary, and returning them unharmed to their habitat where possible.

7 · Mark-scheme phrase bank

Context	Write this	Not this
Why control a variable	"So that any change in the dependent variable is due to the independent variable alone and the results are valid "	"To make it a fair test"
Why a control experiment	"To show the effect is caused by the extract and not the solvent , giving a baseline for comparison"	"To compare"
Why repeat	"To identify anomalous results and calculate a mean , reducing the effect of random error "	"To make it more accurate"
Why initial rate	" Substrate concentration is highest and not yet limiting, so the rate is maximal; it falls as substrate is used up"	"Because it's at the start"
Why percentage change	"Because the initial masses differ , so percentage change allows a fair comparison "	"It's easier to compare"
Overlapping error bars	"The difference is probably not significant and may be due to chance"	"There is no difference"
Non-overlapping error bars	"The difference is likely to be significant ; a statistical test would be needed to confirm it"	"This proves the difference is real"
Large standard deviation	"The data are widely spread about the mean, so the mean is less reliable "	"The results are bad"
Graticule calibration	"Calibrate the eyepiece graticule against a stage micrometer ; the calibration is valid only for that objective lens "	"Use the graticule to measure"
An association in data	"This is a correlation ; it does not establish causation "	"This shows X causes Y"
Describing a trend	Quote figures with units and name the shape — "rises from 2.1 to 6.8 between 0 and 40 minutes, then plateaus"	"It goes up then levels off"
Suggesting an improvement	A specific change plus why it helps — "use a colorimeter, removing the subjective judgement of the end point"	"Be more careful"

8 · The mistakes that cost the grade

1. Rushing the graph. It is four marks against named criteria and deserves six minutes.
2. Axes the wrong way round — **independent on x, dependent on y**.
3. Omitting **units** from axis labels or table headings.
4. Joining points dot-to-dot instead of drawing a **line of best fit**.
5. Drawing a bar chart with no gaps, or a histogram with gaps.
6. No **key** when two data sets are plotted.
7. A non-linear or interrupted scale where one is not permitted.
8. Putting units beside every value in a table instead of in the heading only.
9. Answering "why control this variable" with "to make it a fair test" and nothing more.
0. Confusing a **control variable** with a **control experiment**.
1. Describing a method in vague prose instead of numbered steps with quantities and apparatus.
2. Omitting the quantities — "add some solution" rather than "add 10 cm³".
3. Forgetting to **blot tissue dry** before weighing, or not saying so.
4. Giving raw change in mass where **percentage change** is required, and not explaining why.
5. Saying overlapping error bars mean "no difference" rather than "probably not significant".
6. Claiming a statistical test **proves** a result.
7. Copying a full calculator display instead of rounding sensibly.
8. Dividing by the final value in a percentage change.
9. Using the diameter where the formula needs the radius.
0. Not converting units before a magnification calculation.
1. Listing limitations without pairing each with a **specific** improvement.
2. Extrapolating a conclusion beyond the range of data tested.
3. In a risk assessment, naming a hazard but not the precaution.
4. Leaving a "suggest" question blank because the context was unfamiliar — it is meant to be.

5. Treating the three questions as though the later ones are harder. They are equally weighted.

9 · Final checklist

Can you, from memory...	Can you, from memory...
• State the five graph criteria the mark schemes use? • Decide between a line graph, bar chart, histogram and scatter graph? • Draw a line of best fit correctly, and say why not dot-to-dot? • Lay out a results table with headings, units, repeats and a mean? • Define independent, dependent and control variables? • Give the two-part answer for why a variable is controlled? • Explain the purpose of a control experiment in any context? • Explain why initial rate is measured? • Explain why a particular incubation temperature was chosen? • Distinguish accurate, precise, resolution, reliable and valid? • Say why repeats are done, and what to do with an anomaly?	• Describe making a solution of known concentration in four steps? • Describe a serial dilution with a worked factor? • Give six points of aseptic technique with reasons? • Describe calibrating an eyepiece graticule, and state its limitation? • Describe the plant-tissue osmosis method, including blotting? • Explain why percentage change in mass is used? • Interpret where a line crosses zero on a percentage-change graph? • Do all nine calculation types without hesitating? • Calculate percentage error and say what it implies? • Interpret overlapping and non-overlapping error bars, with the right hedging? • Pair four limitations with four specific improvements? • Give a hazard, its risk and its precaution for four common ones?

THE SHORT VERSION

This paper does not test what you know; it tests whether you think like someone who has actually done the experiment. The marks sit in a small, fixed set of skills that recur in every session — drawing a graph to the criteria, describing a method in quantified steps, explaining why a design decision was made, processing data correctly, and evaluating honestly. None of it depends on recognising the context, so unfamiliarity is never a reason to panic. Work slowly: at 1.6 minutes per mark this is the most generous paper in the qualification, and almost every mark lost on it is lost to haste rather than to ignorance.